Jamilus Sheium

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RESEARCH PROFILE & MATHEMATICAL FOUNDATIONS

First-Principles Machine Learning Architect for Adversarial Security & Privacy — My core expertise lies in designing custom neural architectures, custom pre-training objectives, and mechanistic alignment controls from mathematical first principles, and translating those foundations into high-stakes cybersecurity and privacy systems. Across three externally funded European and UKRI grants, domain-expert PIs recruited me as the sole Lead ML Researcher to architect the underlying mathematical and learning systems:

- **Custom Model & System-1 Architecture Design:** End-to-end design of custom-pretrained Foundation Models, fast **System-1 joint-embedding decision heads**, dilated **Temporal Convolutional Networks (TCNs)**, and **Reinforcement Learning from Contrastive/Contextual Decisions (RLCD)**.
- **First-Principles Neural Mechanics:** Deep mathematical grounding in transformer latent geometry, low-rank subspace projections (refusal directions), quantization error propagation, loss-landscape curvature, and parameter-efficient continual adaptation (LoRA/PEFT).
- **Translation to Privacy & Cybersecurity:** Formulating mathematical ML solutions for multi-institutional **Federated Learning**, **encrypted-traffic sequence representation** (without payload decryption), **client-side semantic Data Loss Prevention (DLP)**, and **out-of-distribution (OOD) zero-day threat detection**.

EDUCATION

University of Portsmouth Portsmouth, UK
BSc (Hons) Computer Science *2025 – 2027 (Expected)*

- **Academic Standing:** First Class Honours — **85% weighted average** across all modules completed to date (highest UK classification band; equivalent to a **4.0/4.0 US GPA**).
- **Honours Dissertation:** *"Alignment Under Evolution: Measuring and Preserving Corrigibility in Recursive Self-Improving LLM Agents."* Formulating measurable indicators of recursive alignment drift and an external provenance-constrained update gate across multi-generation self-modifying agents (scaffolds & PEFT/LoRA).
- **Departmental Distinction:** Sole undergraduate in the School of Computing appointed as Lead ML Research Assistant across three externally funded European and UKRI research grants.

Notre Dame College Dhaka, Bangladesh
Higher Secondary Certificate (Science) *GPA: **5.00 / 5.00** (Top 1% Nationally; Government Merit Scholarship)*

PUBLICATIONS

Under Review

- **Sheium, J.**, Ghahremani, M., & Bader-El-Den, M. (2026). "When Does Post-Training Quantisation Harm LLM Safety? A Pre-Registered Characterisation and the Measurement Pitfalls That Obscure It." Submitted to *IEEE Access* (Under Review). [**First Author**]
  - Conducted a pre-registered, multi-seed, mechanism-aware characterisation of how 4-bit post-training quantisation (PTQ) perturbs internal safety representations across three instruction-tuned model families, four PTQ recipes, and three languages.
  - Derived a layer-wise, **refusal-direction-guided mixed-precision allocation policy** that protects critical safety subspaces under strict 4-bit memory budgets, while mathematically isolating evaluation variance and measurement pitfalls that confound prior work.

RESEARCH & CUSTOM ML ARCHITECTURE EXPERIENCE

Co-Founder & ML Lead — Voxlith (University Spin-Out from UKRI Locknest) *Oct 2025 – Present*
*University of Portsmouth & UKRI CyberASAP | PI: Prof. Stavros Shiaeles | Venture Site: **voxlith.com***
*Originated as the UKRI CyberASAP **Locknest** grant (Oct 2025 – Feb 2026) before spinning out as **Voxlith**, currently deployed across **two active enterprise pilots** (proprietary architectures and weights under commercial NDA).*

- **Custom Cyber Foundation Model & Pre-Training:** Designed and pre-trained a domain-specific Cyber Foundation Model from scratch over a metric-balanced, heterogeneous multi-modal substrate (exceeding 2M structural DOM trees, URLs, HTTP headers, and runtime execution traces) to learn invariant representations of adversarial web behaviour.
- **System-1 Decision Head, TCNs & RLCD:** Architected a low-latency **System-1 joint-embedding decision head** coupled with dilated **Temporal Convolutional Networks (TCNs)** and **Reinforcement Learning (RLCD)**, enabling sub-millisecond, reflex-level zero-day threat containment against adaptive adversaries.
- **End-to-End Ablation & Subspace Optimization:** Formulated the custom loss functions, architecture search, and feature ablation programme from first principles, achieving **96% accuracy** on strictly held-out evaluation splits and transitioning the neural engine into two commercial pilot deployments.

- **Kit-Disjoint OOD Generalisation Protocol:** Proved mathematically and empirically that standard IID random splits suffer severe representation leakage across shared attack-kit templates; formulated a **kit-family-disjoint splitting methodology** to measure true out-of-distribution (OOD) generalisation.

Lead ML Research Assistant — XTRUST-6G (Horizon Europe / SNS JU) *Jul 2026 – Present*
University of Portsmouth | Supervisor: Prof. Stavros Shiaeles
An €8M Horizon Europe consortium developing AI-driven, zero-trust security for 6G networks across 18+ European institutions.

- **Custom Deep Sequence Anomaly Architectures:** Lead the consortium's ML detection workstream, designing deep neural architectures that learn structural and temporal flow manifolds to detect zero-day network intrusions without static signatures.
- **Federated Optimization Privacy–Utility Theory:** Architect multi-institutional **Federated Learning** aggregation protocols enabling distributed 6G nodes to collaboratively optimize shared intrusion detectors under non-IID traffic distributions without raw data exchange, empirically characterising the privacy–utility Pareto frontier.
- **Encrypted-Flow Representation Learning:** Spearhead a new research direction extracting predictive neural representations directly from encrypted session dynamics and packet-timing metadata (bypassing payload decryption), authoring the consortium's evaluation and data schema.
- **Real-Time Streaming Inference Engine:** Engineered low-latency PyTorch/Linux streaming inference pipelines to execute custom sequence models on live high-throughput network telemetry across partner testbeds.

Lead ML Research Assistant — PromptGuard (UKRI CyberASAP) *Mar 2026 – Jul 2026*
University of Portsmouth | Supervisor: Dr. Bander Alrimy | Project Site: ***promptguard-cyberasap.github.io***
UKRI-funded commercialisation project developing local, client-side semantic Data Loss Prevention (DLP) and prompt sanitisation.

- **Contextual Semantic Privacy Formulation:** Formulated sensitive disclosure detection as a deep semantic entailment and latent-representation classification problem rather than rigid token/NER matching, enabling detection of implicit confidential leaks in LLM prompts.
- **Quantised Edge SLM Design Serving:** Adapted and compressed Small Language Models (SLMs) deployed via local inference engines (**vLLM / Ollama**) to operate within strict client-side memory and latency bounds, applying mechanistic subspace insights from concurrent PTQ research.
- **Synthetic Benchmark Leakage-Free Evaluation:** Designed a multi-stage dataset synthesis and gold-set annotation protocol to evaluate semantic privacy preservation fairly against commercial NER/DLP baselines.

HONOURS & ACADEMIC DISTINCTIONS

- **21st Place Nationally — BUET Admission Examination:** Ranked 21st across Bangladesh in the undergraduate entrance examination for the Bangladesh University of Engineering and Technology (BUET), the country's most selective engineering and mathematics examination.
- **Commercial Spin-Out & Live Enterprise Pilots (Voxlith):** Co-founded University of Portsmouth spin-out (voxlith.com) following UKRI CyberASAP validation, deploying custom Foundation Model + System-1 TCN architectures across two active commercial pilots.
- **3x Competitive Undergraduate ML Lead Appointments:** Recruited as the lead mathematical and ML architect across three consecutive externally funded grants (Horizon Europe & UKRI CyberASAP) during undergraduate studies.

TECHNICAL SKILLS

- **Neural Architecture Mathematical Foundations:** Custom Foundation Model pre-training; System-1 joint-embedding decision heads; Temporal Convolutional Networks (TCNs); RLCD; low-rank subspace geometry refusal directions; Post-Training Quantisation (PTQ) mixed-precision allocation; Federated optimization; PEFT/LoRA continual adaptation.
- **Privacy, Security Evaluation Methodology:** Pre-registered experimental design; recursive agentic corrigibility update gating; kit-disjoint OOD splitting; encrypted-traffic representation learning; semantic DLP; variance decomposition.
- **Compute, Frameworks Systems:** PyTorch, TensorFlow/Keras, HuggingFace Transformers; NVIDIA L40S multi-GPU training orchestration; vLLM, Ollama, CUDA, Docker, Linux; Python, C, TypeScript/JavaScript, PostgreSQL, Bash.

REFERENCES

- **Prof. Stavros Shiaeles** — Professor in Cybersecurity, University of Portsmouth *(PI: XTRUST-6G & Locknest / Voxlith)*
- **Dr. Bander Alrimy** — Senior Lecturer in Cybersecurity, University of Portsmouth *(PI: PromptGuard)*
- **Dr. Mani Ghahremani** — Lecturer in Computing, University of Portsmouth *(Co-Author & Research Supervisor)*